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Codeword or Noise? Exact Random Coding Exponents for Slotted Asynchronism*

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Abstract

We consider the problem of slotted asynchronous coded communication, where in each time frame (slot), the transmitter is either silent or transmits a codeword from a given (randomly selected) codebook. The task of the decoder is to decide whether transmission has taken place, and if so, to decode the message. We derive the optimum detection/decoding rule in the sense of the best trade-off among the probabilities of decoding error, false alarm, and misdetection. For this detection/decoding rule, we then derive single–letter characterizations of the exact exponential rates of these three probabilities for the average code in the ensemble.

Index Terms: Synchronization, error exponent, false alarm, misdetection, random coding.

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